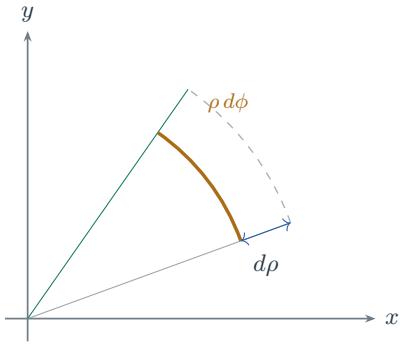
曲线坐标：坐标增量先换成真实长度



$$d\mathbf{r} = \sum_i h_i\, dq_i\, \mathbf{e}_i$$

$h_i dq_i$ 才是真实边长。

$$\nabla f = \sum_i \frac{1}{h_i} \frac{\partial f}{\partial q_i} \mathbf{e}_i$$

柱坐标：$(h_\rho, h_\phi, h_z) = (1, \rho, 1)$

球坐标：$(h_r, h_\theta, h_\phi) = (1, r, r\sin\theta)$